EXERCÍCIOS - PORTAS LÓGICAS

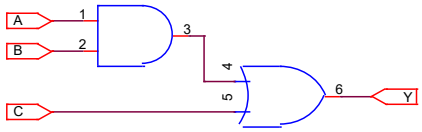
1. Escreva uma expressão Booleana que seja 0 apenas quando todas as suas variáveis de entrada (A, B, C e D) forem 0s.

2. Escreva uma expressão Booleana que seja 1 quando uma ou mais de suas variáveis de entrada (A, B, C, D e E) forem 0s.

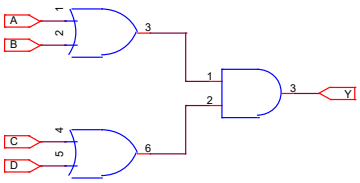
3. Escreva uma expressão Booleana que seja 0 quando uma ou mais de suas variáveis de entrada (A, B, C, D e E) forem 0s.

4. Apresente a função booleana e a tabela verdade dos circuitos digitais apresentados abaixo:

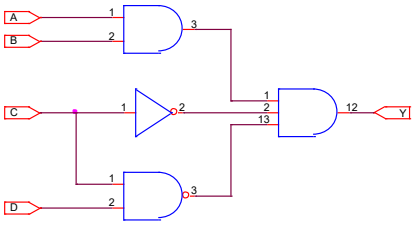
a.



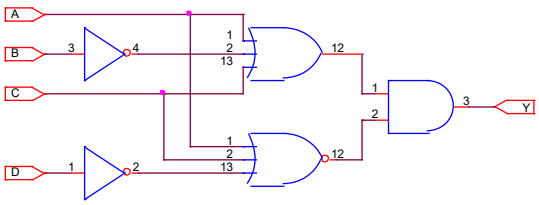
b.



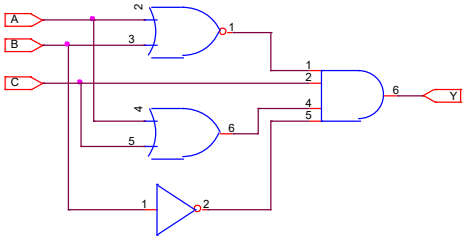
c.



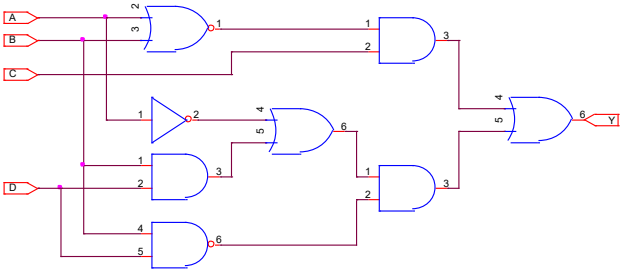
d.



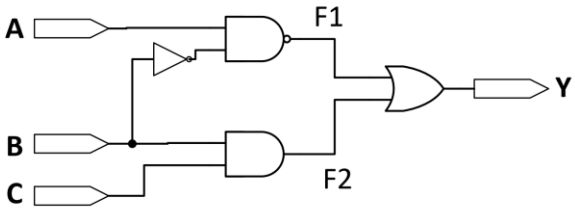
e.



f.

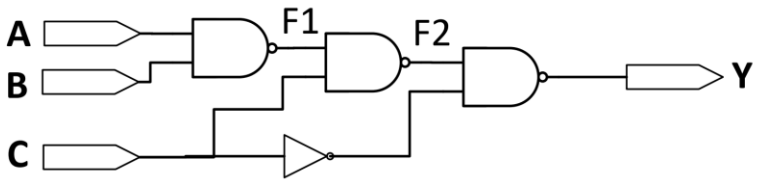


5. Considerando os diagramas esquemático abaixo, pede-se: equação booleana e tabela verdade.



A	B	C	F1	F2	Y
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

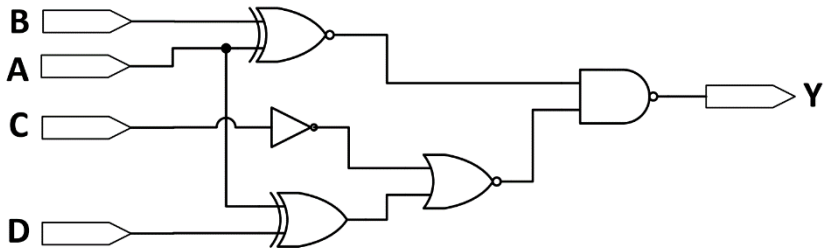
Y = _____



A	B	C	F1	F2	Y
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

Y = _____

Considerando o diagrama esquemático abaixo, pede-se: equação booleana.



Y = _____

6. Desenhe as expressões Booleanas apresentadas abaixo:

a. $f1 = \overline{A}.B + \overline{AB}$
b. $f2 = A.B + C$
c. $f2 = A.\overline{B} + A.B.\overline{C}$
d. $f4 = Z.\left(X + \overline{XY}\right)$
e. $f5 = (A + B + C).\left(\overline{A} + \overline{B} + C\right)$
f. $f6 = A \odot (\overline{B} + (\overline{(\overline{A.B}) + (C \oplus D)}))$

7. Apresente o diagrama esquemático e a tabela verdade das funções booleanas apresentadas abaixo:

a. $f(A,B,C) = (A + B) \cdot \overline{C}$
b. $f(A,B,C) = (A \cdot B) + \left(A \cdot \overline{C}\right)$
c. $f(A,B,C) = \left(A \cdot B \cdot \overline{C}\right) + \overline{(A \oplus C)}$
d. $f(A,B,C) = (B \cdot C) + \left(\overline{A} \cdot \overline{C}\right)$
e. $f(A,B,C) = \left(A \cdot \overline{B}\right) + (A \cdot B \cdot \overline{C}) + (A \cdot C)$
f. $f(A,B,C,D) = \overline{\left(A \cdot \overline{B}\right) + \left(\overline{C \cdot D}\right)}$
g. $f(A,B,C,D) = \overline{\left((A \cdot B \cdot \overline{C}) + \overline{(C \cdot \overline{D})}\right) \oplus D}$
h. $f(A,B,C,D,E,F) = \overline{\left(\left(\left(A \cdot \overline{B} \cdot C\right) + D\right) \oplus E\right) \cdot \overline{F}}$

8. Fazer a tabela verdade para as seguintes funções e desenhar o circuito equivalente:

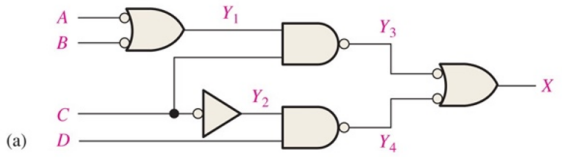
$F1 = \bar{A}.B + A.B.\bar{C} + \bar{A}.\bar{C} + A.\bar{B}.C$

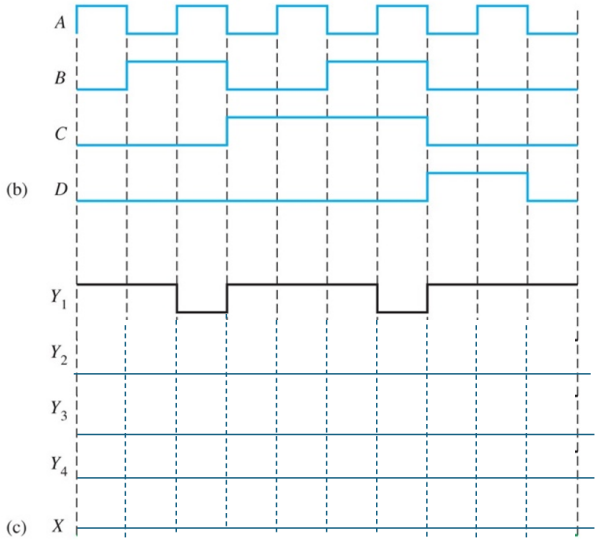
A	B	C	F1
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

$F2 = \bar{A} + B.\bar{C} + C.D + A.\bar{B}.C$

A	B	C	D	F2
0	0	0	0	
0	0	0	1	
0	0	1	0	
0	0	1	1	
0	1	0	0	
0	1	0	1	
0	1	1	0	
0	1	1	1	
1	0	0	0	
1	0	0	1	
1	0	1	0	
1	0	1	1	
1	1	0	0	
1	1	0	1	
1	1	1	0	
1	1	1	1	

9. Determine as formas de onda para os sinais Y2 a Y4, assim para a saída X:





$\boldsymbol{Y1 = \bar{A} + \bar{B}}$

10. Mostre o resultado das seguintes operações bit a bit (todos os valores estão representados em hexadecimal):

```
a. not 0x99
b. not 0x01
c. 0x99 and 0x99
d. 0x99 and 0xFF
e. 0x99 or 0x99
f. 0x99 or 0xff
g. not (0x99 or 0x99)
```

11. Qual o resultado das seguintes operações bit-a-bit? Assuma que os valores (em decimal) estejam representados em complemento de 2 (8 bits)

```
a. 22 and 5
b. not -1
c. -5 or 7
d. -10 xor 8
e. 66 and 59
f. 98 or -12
```

12. Apresente máscaras binárias e a respectiva operação para os seguintes casos (valores de 8 bits)

a. Limpar (forçar para 0) os quatro bits mais significativos
b. Marcar (forçar para 1) os quatro bits menos significativos
c. Inverter os três bits menos significativos e os dois mais significativos
d. Marcar o terceiro bit menos significativo e limpar o bit mais significativo

13. Apresente as funções boolenas e os diagramas esquemáticos dos circuitos digitais dados nas tabelas verdade abaixo:

a.

A	B	C	Y
0	0	0	**1**
0	0	1	**1**
0	1	0	**0**
0	1	1	**1**
1	0	0	**0**
1	0	1	**1**
1	1	0	**1**
1	1	1	**1**

b.

A	B	C	D	Y
0	0	0	0	**1**
0	0	0	1	**1**
0	0	1	0	**0**
0	0	1	1	**1**
0	1	0	0	**0**
0	1	0	1	**1**
0	1	1	0	**1**
0	1	1	1	**1**
1	0	0	0	**1**
1	0	0	1	**1**
1	0	1	0	**0**
1	0	1	1	**1**
1	1	0	0	**0**
1	1	0	1	**1**
1	1	1	0	**0**
1	1	1	1	**1**

c.

A	B	C	D	Y
0	0	0	0	**0**
0	0	0	1	**0**
0	0	1	0	**0**
0	0	1	1	**0**
0	1	0	0	**1**
0	1	0	1	**1**
0	1	1	0	**0**
0	1	1	1	**0**
1	0	0	0	**0**
1	0	0	1	**0**
1	0	1	0	**0**
1	0	1	1	**0**
1	1	0	0	**1**
1	1	0	1	**1**
1	1	1	0	**0**
1	1	1	1	**0**

d.

A	B	C	D	Y
0	0	0	0	**1**
0	0	0	1	**0**
0	0	1	0	**1**
0	0	1	1	**0**
0	1	0	0	**0**
0	1	0	1	**0**
0	1	1	0	**0**
0	1	1	1	**0**
1	0	0	0	**1**
1	0	0	1	**0**
1	0	1	0	**1**
1	0	1	1	**0**
1	1	0	0	**0**
1	1	0	1	**0**
1	1	1	0	**0**
1	1	1	1	**0**

e.

A	B	C	D	Y
0	0	0	0	**0**
0	0	0	1	**0**
0	0	1	0	**0**
0	0	1	1	**0**
0	1	0	0	**1**
0	1	0	1	**1**
0	1	1	0	**1**
0	1	1	1	**1**
1	0	0	0	**0**
1	0	0	1	**0**
1	0	1	0	**0**
1	0	1	1	**0**
1	1	0	0	**0**
1	1	0	1	**0**
1	1	1	0	**0**
1	1	1	1	**0**

f.

A	B	C	D	Y
0	0	0	0	**1**
0	0	0	1	**1**
0	0	1	0	**1**
0	0	1	1	**1**
0	1	0	0	**0**
0	1	0	1	**0**
0	1	1	0	**0**
0	1	1	1	**0**
1	0	0	0	**1**
1	0	0	1	**1**
1	0	1	0	**1**
1	0	1	1	**1**
1	1	0	0	**0**
1	1	0	1	**0**
1	1	1	0	**0**
1	1	1	1	**0**

g.

A	B	C	D	Y
0	0	0	0	**0**
0	0	0	1	**1**
0	0	1	0	**0**
0	0	1	1	**1**
0	1	0	0	**0**
0	1	0	1	**0**
0	1	1	0	**0**
0	1	1	1	**0**
1	0	0	0	**0**
1	0	0	1	**1**
1	0	1	0	**0**
1	0	1	1	**1**
1	1	0	0	**0**
1	1	0	1	**0**
1	1	1	0	**0**
1	1	1	1	**0**

h.

A	B	C	D	Y
0	0	0	0	**1**
0	0	0	1	**0**
0	0	1	0	**0**
0	0	1	1	**1**
0	1	0	0	**1**
0	1	0	1	**1**
0	1	1	0	**1**
0	1	1	1	**1**
1	0	0	0	**0**
1	0	0	1	**1**
1	0	1	0	**0**
1	0	1	1	**0**
1	1	0	0	**1**
1	1	0	1	**0**
1	1	1	0	**1**
1	1	1	1	**1**

i.

A	B	C	D	Y
0	0	0	0	**1**
0	0	0	1	**1**
0	0	1	0	**0**
0	0	1	1	**0**
0	1	0	0	**1**
0	1	0	1	**1**
0	1	1	0	**0**
0	1	1	1	**0**
1	0	0	0	**1**
1	0	0	1	**1**
1	0	1	0	**0**
1	0	1	1	**0**
1	1	0	0	**1**
1	1	0	1	**1**
1	1	1	0	**0**
1	1	1	1	**0**